MAS508-G Control Theory
Lecture 3: State Space Design

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L3. State Space Design

a) Time Response
b) Controllability
c) Observability
d) Pole Placement Control
e) Observer Design
f) Output Feedback Control
g) Optimal Control - Linear Quadratic Control (LQR)
h) Optimal Control – LQG

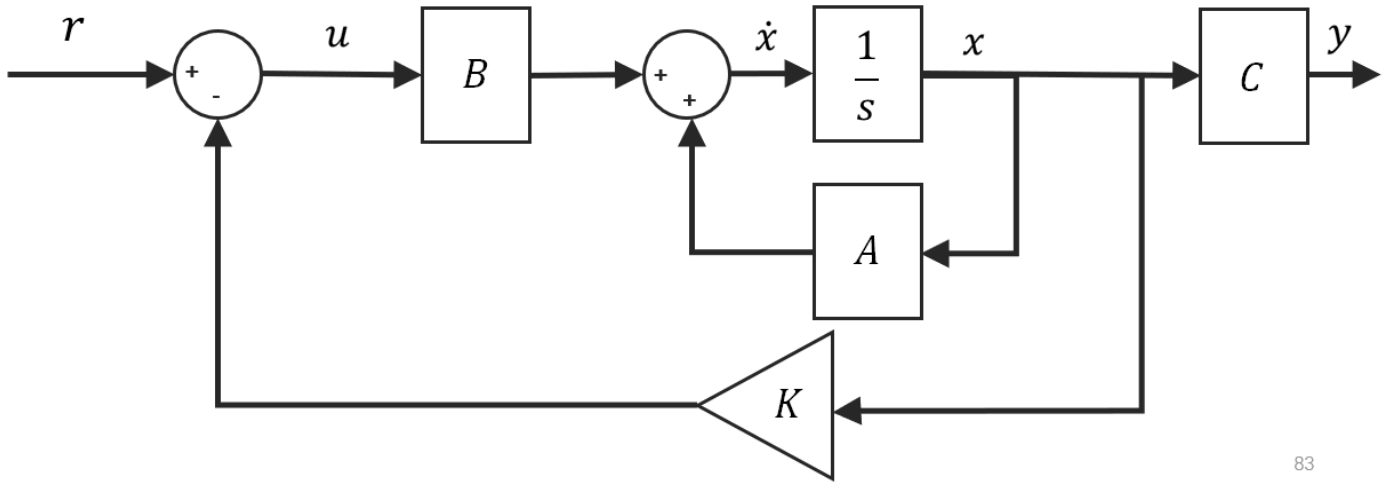
Textbook, “Control Systems Engineering”, Chapters 4 &12

L3d. State Space Control - Pole Placement Control

Book “Control Systems Engineering”, Chapter 12.2, 12.4

What is Pole Placement?

- Pole placement is a *state-feedback control method*
- System $\dot{x} = Ax + Bu, \quad y = Cx$
  State feedback control: $u = -\boldsymbol{K}x + r$
- It calculates the control gain you need to place the poles of system to a desired location.



Why Pole Placement?



- One major benefit of modeling a system in state-space is being able to use **state-feedback control**
- State-feedback control allows you to control more complex Multiple-Input Multiple-Output (MIMO) systems.
- Pole placement is a state feedback design method

Pole Placement Design

- For a system with the state-space model:

$$\dot{x} = Ax + Bu, \qquad y = Cx$$

- Pole-placement finds a state-feedback control law:

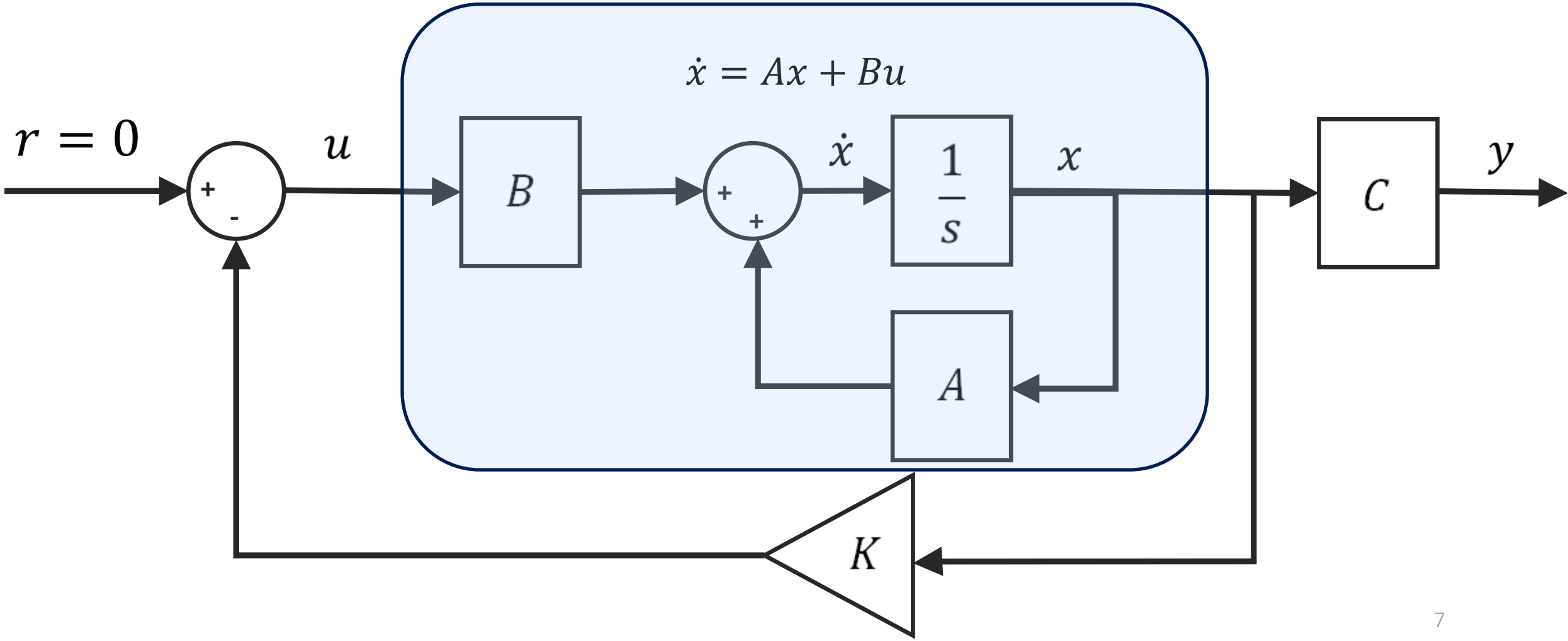
$$u = -Kx$$

- That brings the poles of the closed-loop system to desired locations

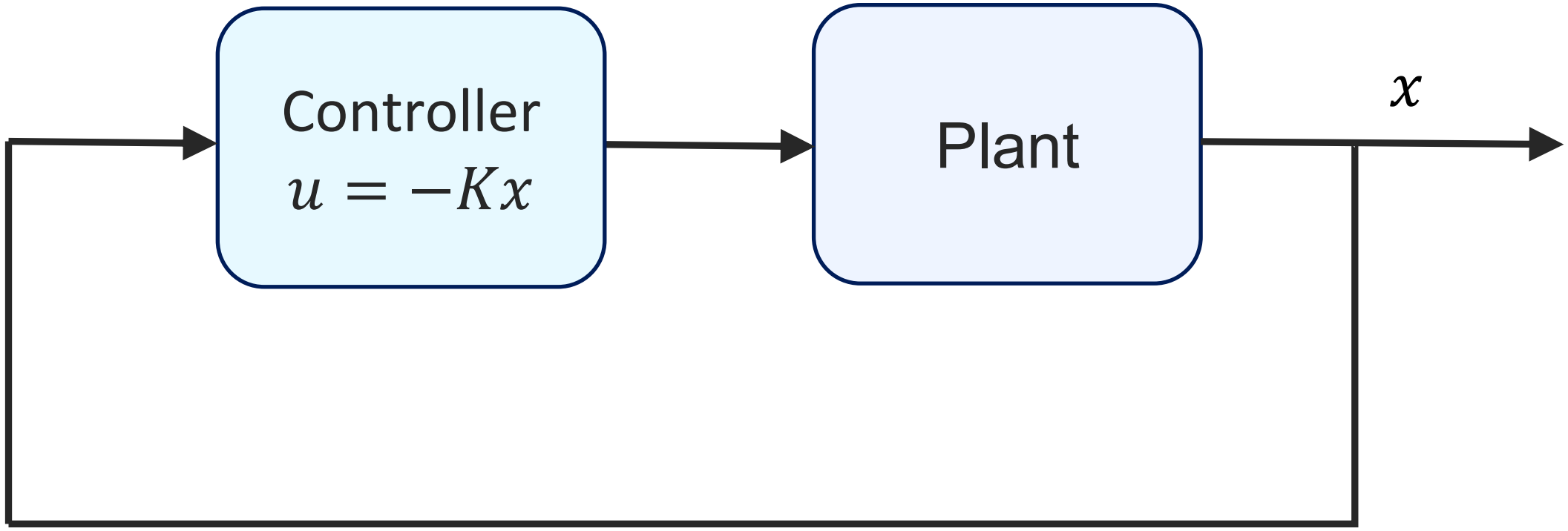
$$\dot{x} = (A - BK)x$$

- Such as the characteristic equation $\mathbf{det}\big(sI - (A - BK)\big) = \mathbf{0}$ matches the desired characteristic equation. (Desired poles)

State Feedback $u = -Kx$



State feedback control



Conditions: Pole Placement Control

1. The poles of a system can only be placed **arbitrarily** via state-feedback if and only if the system is completely state **controllable**

2. All the states are available, i.e. being measured directly.

Note: If the system is stabilizable, then state-feedback can be used to make the entire system stable .

Pole-Placement Design

STEP 1: Check the system is controllable. $\dot{x} = Ax + Bu$

$$C_{ab} = (B \quad AB \quad A^2B \ \cdots \ A^{n-1}B), \ rank(C_{ab}) = n.$$

STEP 2: Apply control law $u = -Kx$ to the system and compute the control gain K that assigns the poles to the desired locations.

- Calculate the desired characteristic equation: $(s - p_1)(s - p_2) \cdots (s - p_n)$.
- Closed-loop system is $\dot{x} = (A - BK)x$.
- Find the characteristic equation for the closed-loop system: $\det(s\boldsymbol{I} - (A - BK))$.
- Find the gain $K = [k_1 \ k_2 \cdots k_n]$ to equate the coefficients of the following.

$$\det(s\boldsymbol{I} - (A - BK)) = (s - p_1)(s - p_2) \cdots (s - p_n).$$

Example 1

Consider the following *state-space* system:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 3 & -1 & -5 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Find gain $K = [k_1 \quad k_2 \quad k_3]$ needed to move its closed-loop poles to:

$$CP = [-1 \quad -2 \quad -3]$$

Solution: Check the system is controllable.

$$Rank(C_{ab}) = Rank([B \quad AB \quad A^2B]) = 3.$$

The desired characteristic equation is

$$(s+1)(s+2)(s+3) = s^3 + 6s^2 + 11s + 6$$

Applying the state-feedback control $u = -Kx$ with the gain $K = [k_1 \quad k_2 \quad k_3]$ the closed-loop system becomes:

$$(A - BK) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 3 - k_1 & -1 - k_2 & -5 - k_3 \end{bmatrix}$$

The characteristic equation of $A - BK$ is

$$s^3 + (k_3 + 5)s^2 + (k_2 + 1)s + (k_1 - 3) = s^3 + 6s^2 + 11s + 6$$

In order to obtain the *desired characteristic equation* we need the gain:

$$k_1 - 3 = 6$$
$$k_2 + 1 = 11$$
$$k_3 + 5 = 6$$

Solve for the gains:

$$K = [9 \quad 10 \quad 1]$$

This is the gain required to move the poles to their desired location.

Pole Placement Method

Recall the state-space model of linear system:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

where:

A = system matrix $(n \times n)$
B = input matrix $(n \times r)$
C = output matrix $(m \times n)$
D = feedforward matrix $(m \times r)$

The system state is

$$x = [x_1 \quad x_2 \quad x_3 \ldots x_{n-1} \quad x_n]^T$$

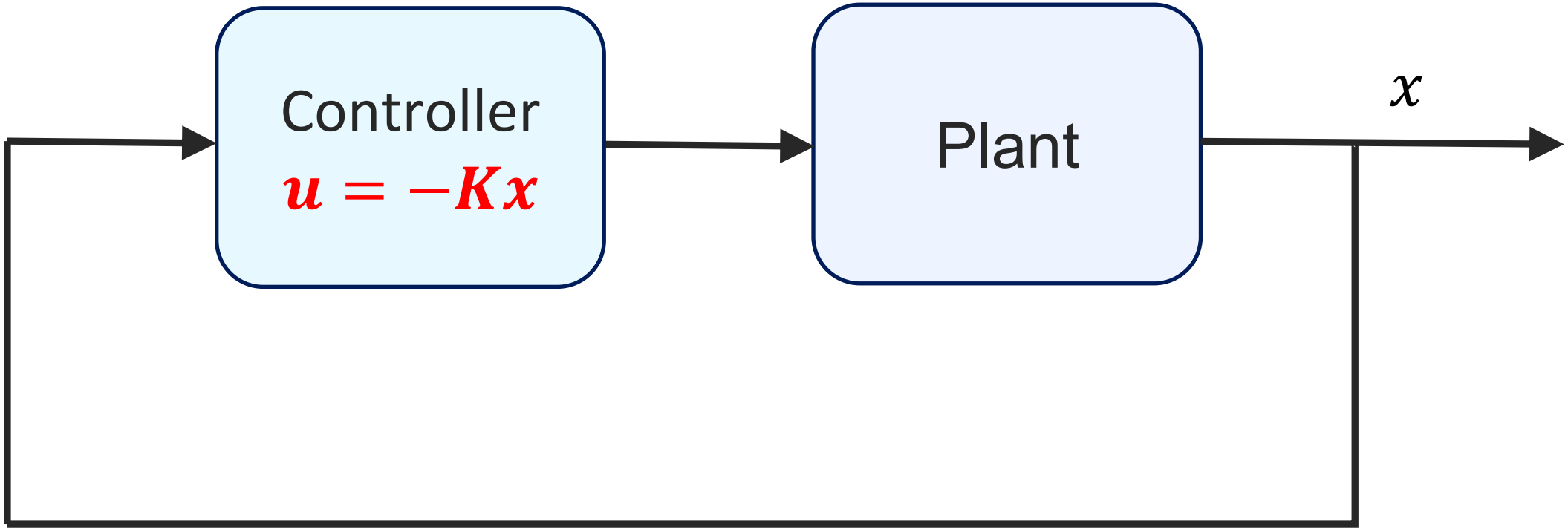
STEP 1: Check the system is controllable. $\dot{x} = Ax + Bu$

$$C_{ab} = (B \quad AB \quad A^2B \quad \cdots \quad A^{n-1}B), \; rank(C_{ab}) = n.$$

STEP 2: Apply control law $\boldsymbol{u = -Kx}$ to the system and compute the control gain K that assigns the poles to the desired locations in the closed-loop system $(A - BK)$.

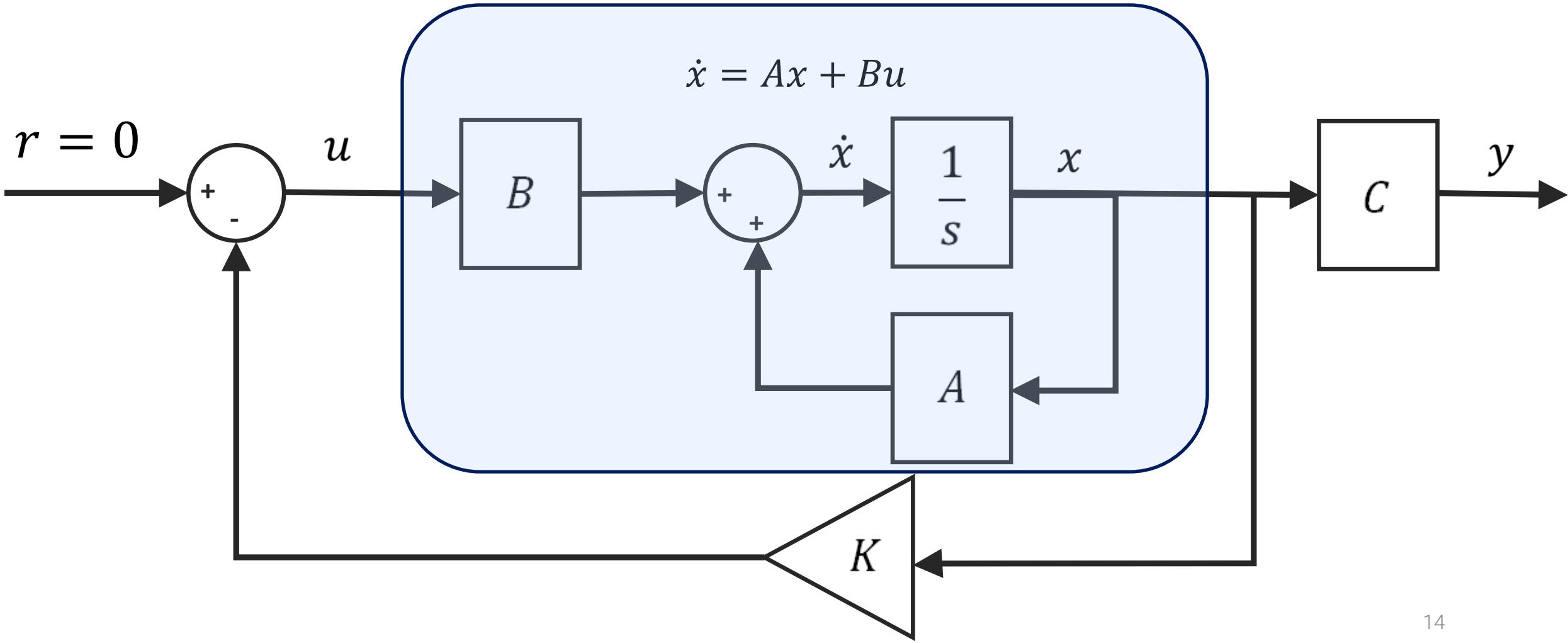
Find the gain $K = [k_1 \; k_2 \cdots k_n]$ to equate the coefficients of $det(s\boldsymbol{I} - (A - BK))$ and $(s - p_1)(s - p_2) \cdots (s - p_n)$.

State feedback control




```
% Matlab
% pole placement control
% P=[p1 p2 . . . ] are  desired poles
% A, B, C, D system state-space matrices
K=place(A,B,P)
```

State Feedback $u = -Kx$



Pole Placement Method

Recall the state-space model of linear system:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

where:

A = system matrix $(n \times n)$
B = input matrix $(n \times r)$
C = output matrix $(m \times n)$
D = feedforward matrix $(m \times r)$

The system state is

$$x = [x_1 \quad x_2 \quad x_3 \ldots x_{n-1} \quad x_n]^T$$

STEP 1: Check the system is controllable. $\dot{x} = Ax + Bu$

$$C_{ab} = (B \quad AB \quad A^2B \quad \cdots \quad A^{n-1}B), \; rank(C_{ab}) = n.$$

STEP 2: Apply control law $\boldsymbol{u = -Kx}$ to the system and compute the control gain K that assigns the poles to the desired locations in the closed-loop system $(A - BK)$.

Find the gain $K = [k_1 \; k_2 \cdots k_n]$ to equate the coefficients of $det(s\boldsymbol{I} - (A - BK))$ and $(s - p_1)(s - p_2) \cdots (s - p_n)$.

Converting the state-space model into the controllable *companion form* makes it easier to do the control design.

Controllable Canonical Form

The controllable canonical form is the form where the characteristic polynomial of the system appears explicitly in the last row of the A matrix.

$$y^{(n)} + a_{n-1}\,y^{(n-1)} + \cdots + a_1\dot{y} + a_0 y = b_{n-1}u^{(n-1)} + b_{n-2}\,u^{(n-2)} + \cdots + b_1\dot{u} + b_0 u$$

$$\frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

$$\dot{x} = A_c x + B_c u$$
$$y = C_c x + D_c u$$

$$A_c = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}$$

$$B_c = [0 \;\; 0 \cdots 0 \;\; 1]^T$$

$$C_c = [b_0 \;\; b_1 \;\; \ldots \;\; b_{n-2} \;\; b_{n-1}]$$

$$D_c = 0$$

Controllable Canonical Form

- The controllable canonical form is the form where the characteristic polynomial of the system appears explicitly in the last row of the A matrix.
- This state-space realization is called controllable canonical form because the resulting model is guaranteed to be controllable
- The controllable canonical form is useful for controller design using pole placement method.
- However, the transformation of the system to companion form is based on the controllability matrix which is almost always numerically singular for mid-range orders. Hence, avoid using the controllable form for computation when possible.

Finding Controllable Companion Form

- Original system $\dot{x} = Ax + Bu \qquad y = Cx$

- State transformation $z = W^{-1}x$ to the controllable companion form

  $\dot{z} = W^{-1}\dot{x} = W^{-1}(Ax + Bu) = W^{-1}AWz + W^{-1}Bu = A_c z + B_c u \ , \ y = CWz$

  $A_c = W^{-1}AWz, \ B_c = W^{-1}Bu$

- The transform matrix is $W = Q\, Q_c^{-1}$
  - Q is Controllability matrix of (A, B) $\quad Q = [B \quad AB \quad A^2B \ \cdots \ A^{n-1}B]$
  - Q_c is Controllability matrix of (A_c, B_c) in controllable companion form

- Design control u for controllable form $\dot{z} = A_c z + B_c u$

Finding Controllability Matrix Q_c

- Original system $\dot{x} = Ax + Bu \qquad y = Cx$
- Find the characteristic equation

  $$\det(sI - A) = s^n + a_{n-1}\, s^{n-1} + \cdots + a_1 s + a_0 = 0$$

- Using the coefficients of characteristic equation to represent the system into the controllable companion form

$$A_c = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \qquad B_c = [0 \;\; 0 \cdots 0 \;\; 1]^T$$

- Calculate $Q_c = [B_c \;\; A_c B_c \;\; A_c^2 B_c \;\cdots\; A_c^{n-1} B_c]$

Pole-Placement Procedure by Transforming to Controllable Companion Form

STEP 1: Find the controllable companion matrices (Ac, Bc) of the system. Find the transformation matrix $W = Q\ Q_c^{-1}$

STEP 2: Apply control law to the controllable companion matrices and compute the companion control gain K_c that assigns the poles to the desired locations in the closed-loop system $(Ac\ Bc\ Kc)$.

STEP 3: Find K=$K_c\ W^{-1}$ to get the feedback gain for the original system (A, B).

Time domain specifications

- For 2nd order system, the time domain specification are percent overshoot, settling time, or peak time (rise time).

- Relation between poles and time domain specifications

Control Specifications for 2nd-order underdamped system

Overshoot of 2nd order system:

$$\zeta = \frac{-\ln(OS)}{\sqrt{\pi^2 + \ln^2(OS)}}$$

T_s settling time of 2nd order system:

$$\omega_n = \frac{4}{\zeta\, T_s}$$

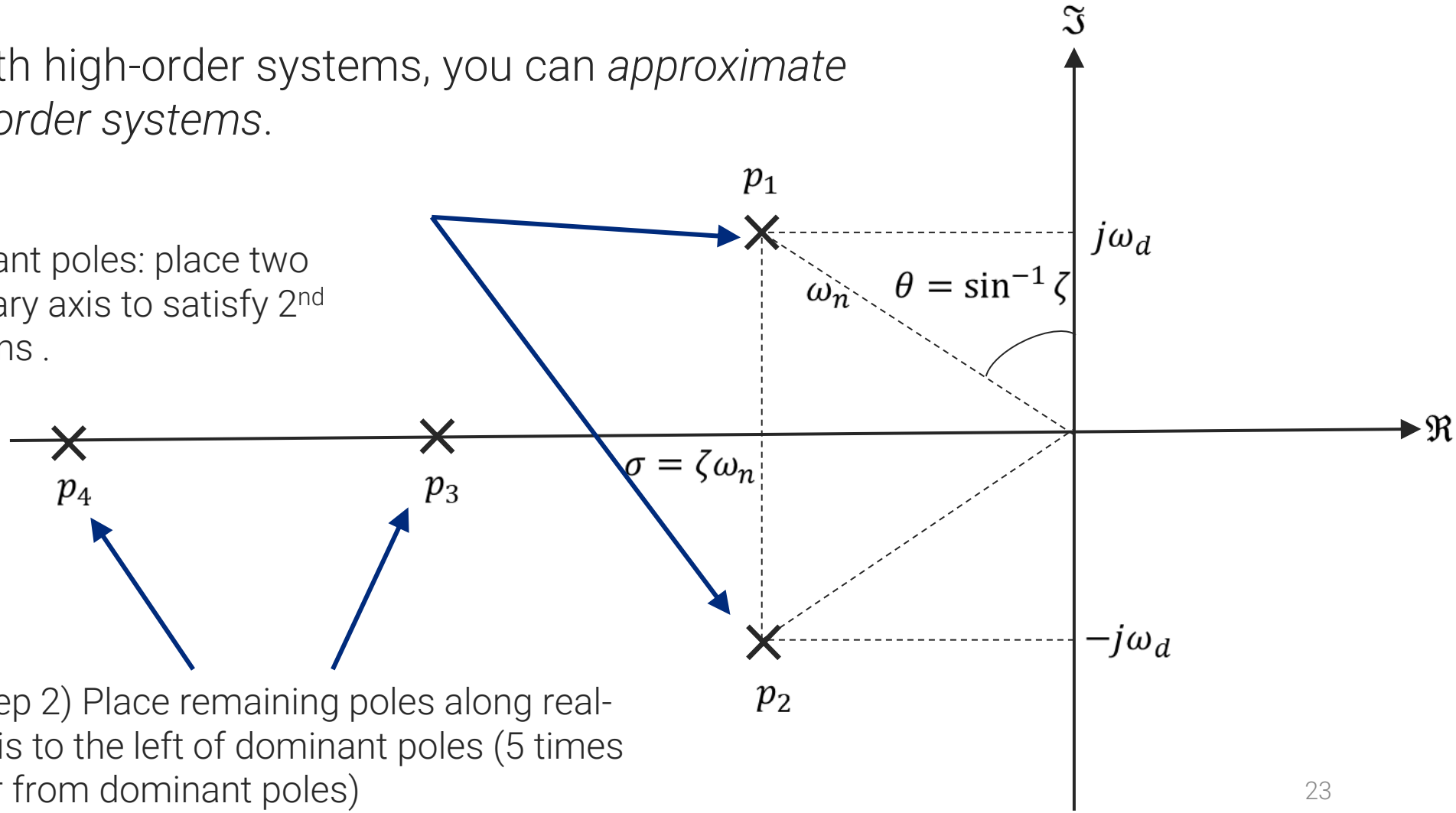
T_p settling time of 2nd order system:

$$\omega_n = \frac{\pi}{\sqrt{1-\zeta^2}\, T_p}$$

Higher-Order System Approximations

When dealing with high-order systems, you can *approximate them as second-order systems*.

Step 1) Select two dominant poles: place two poles close to the imaginary axis to satisfy 2nd order system specifications .



Step 2) Place remaining poles along real-axis to the left of dominant poles (5 times far from dominant poles)

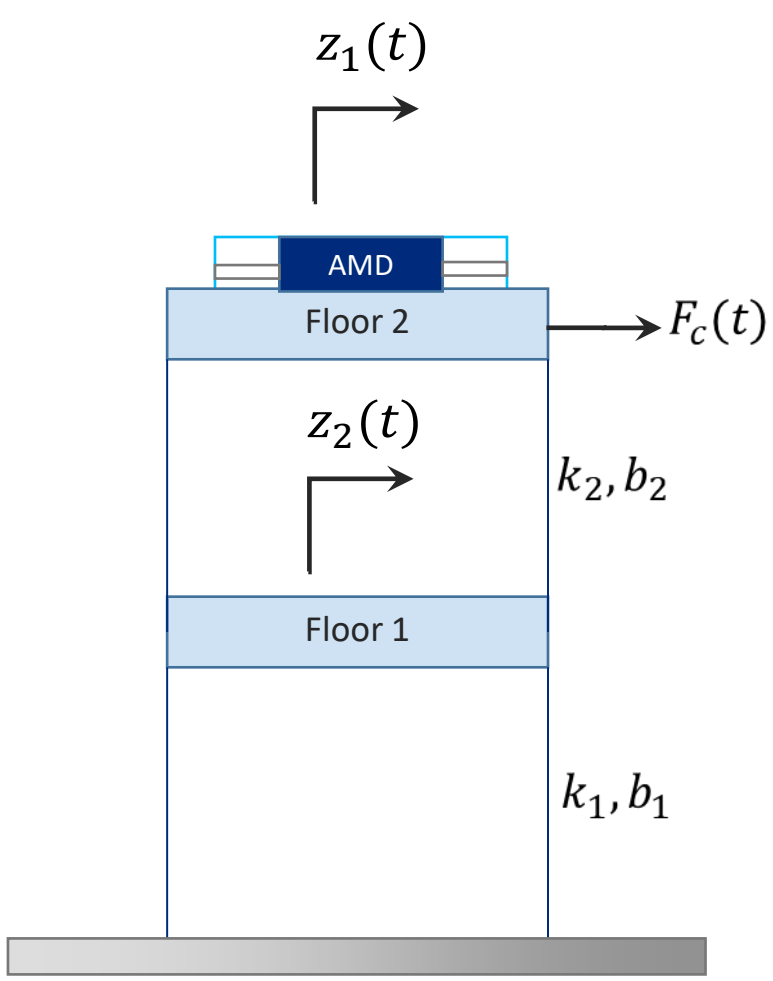
Testing Approximation

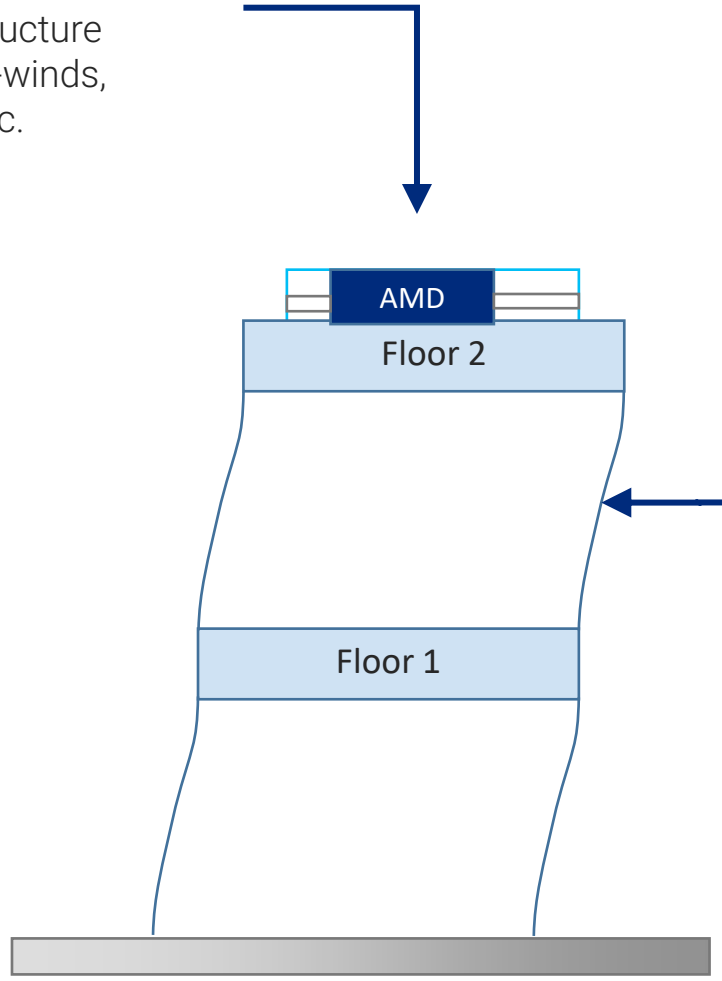
- Closed-loop system should be simulated prior to implementation to ensure:
  - Control requirements are met
  - Control effort is not too high.

- For example, if the poles, p_3 and p_4, are placed too far in the left-hand plane then the control effort may be very high and not realizable in the actual system.

Example 2 Two-Floor Structure with Active Damping

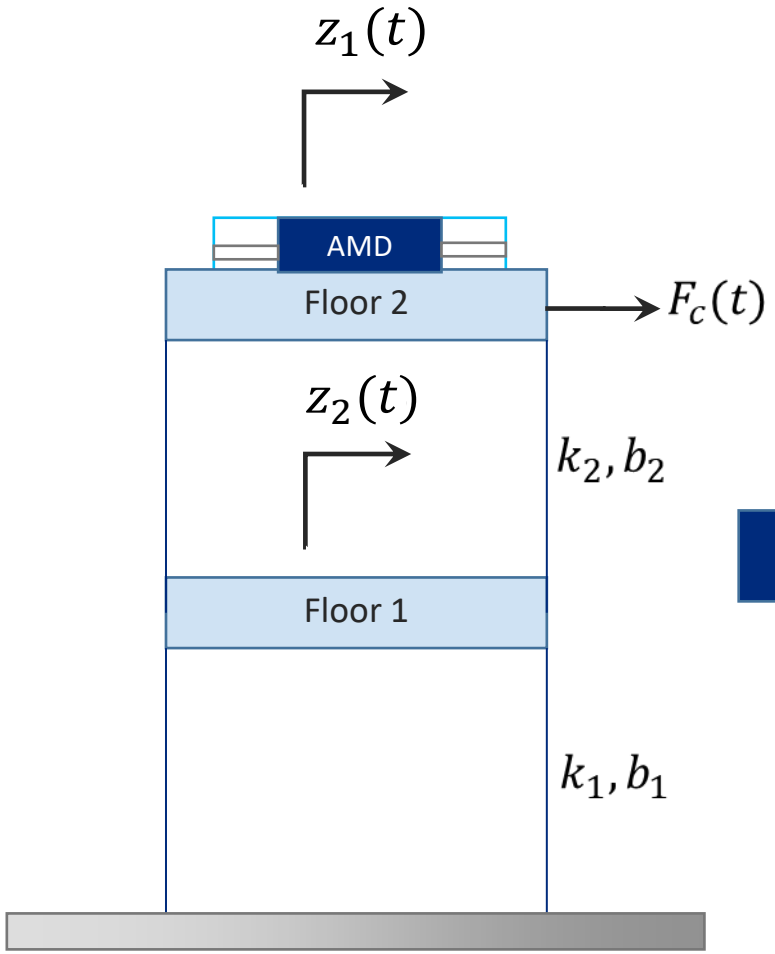
Active Mass Damper (AMD) moves to minimize deflection in structure caused by high-winds, earthquakes, etc.





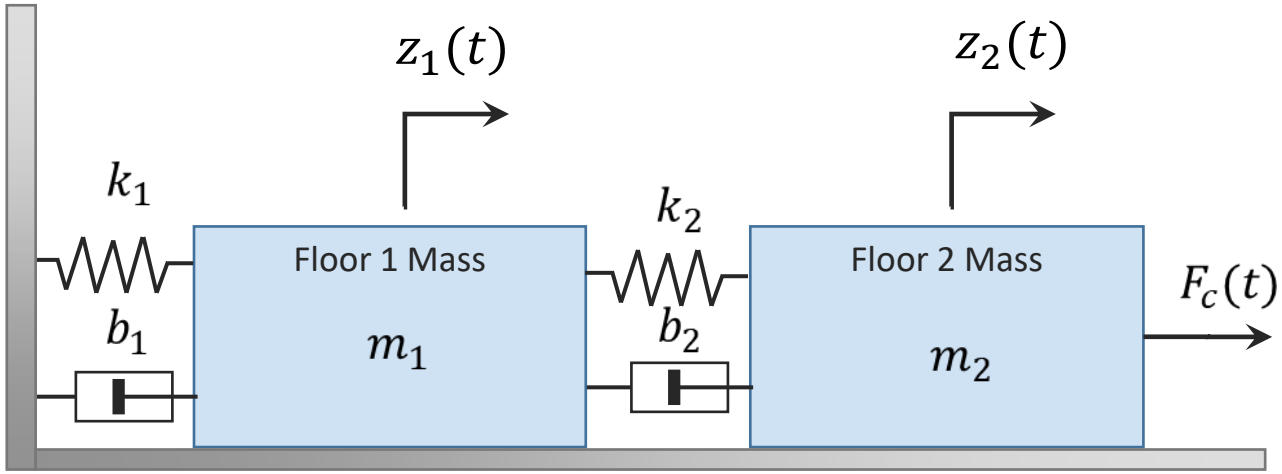
Structure exhibits flexibilities and can be modeled as spring-mass-damper

Example 2 Modelling



Equations of motion:

$$m_1\ddot{z}_1 = k_2(z_2 - z_1) + b_2(\dot{z}_2 - \dot{z}_1) - k_1 z_1 - b_1\dot{z}_1$$
$$m_2\ddot{z}_2 = -k_2(z_2 - z_1) - b_2(\dot{z}_2 - \dot{z}_1) + F_c$$



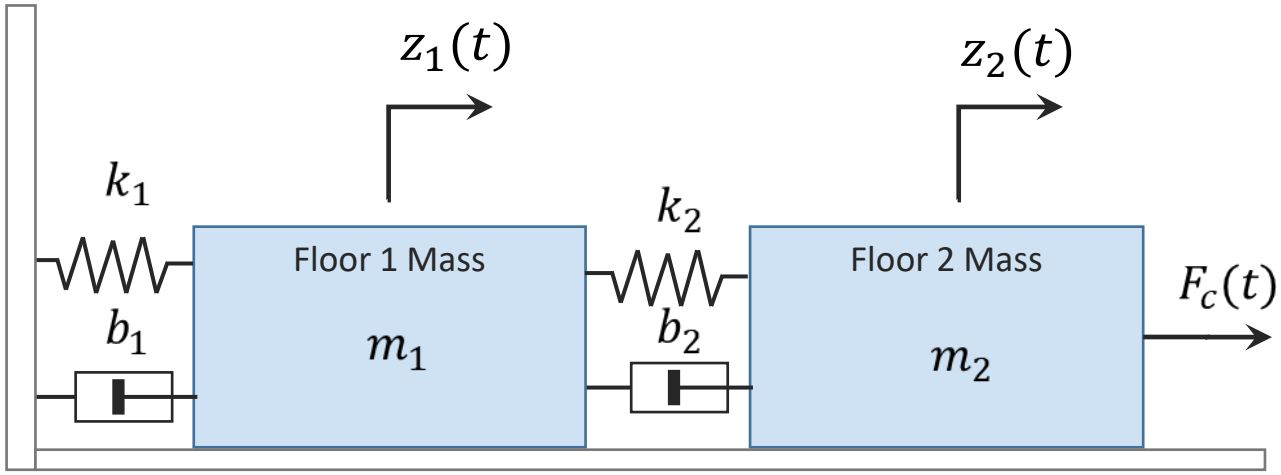
Only measuring the floor accelerations:

$$y = [\ddot{z}_1 \quad \ddot{z}_2]$$

Example 2 State-Space Representation

Equations of motion:

$$m_1\ddot{z}_1 = k_2(z_2 - z_1) + b_2(\dot{z}_2 - \dot{z}_1) - k_1 z_1 - b_1\dot{z}_1$$
$$m_2\ddot{z}_2 = -k_2(z_2 - z_1) - b_2(\dot{z}_2 - \dot{z}_1) + F_c$$



Substituting the state variable $x = [z_1 \;\; z_2 \;\; \dot{z}_1 \;\; \dot{z}_2]$ and control input $u = F_c$:

$$x_1 = z_1, x_2 = z_2, x_3 = \dot{z}_1, x_4 = \dot{z}_2$$

$$m_1\dot{x}_3 = k_2(x_2 - x_1) + b_2(x_4 - x_3) - k_1 x_1 - b_1 x_2$$
$$m_2\dot{x}_4 = -k_2(x_2 - x_1) - b_2(x_4 - x_3) + u$$

Collecting the coefficients:

$$m_1\dot{x}_3 = -(k_1 + k_2)x_1 + k_2 x_2 - (b_1 + b_2)x_3 + b_2 x_4$$
$$m_2\dot{x}_4 = k_2 x_1 - k_2 x_2 + b_2 x_3 - b_2 x_4 + u$$

Output $y = [\ddot{z}_1 \quad \ddot{z}_2] = [\dot{x}_3 \quad \dot{x}_4]$

Example 2 Linear State-Space Model

$$\dot{x}_1 = x_3$$
$$\dot{x}_2 = x_4$$
$$\dot{x}_3 = \frac{1}{m_1}(-(k_1+k_2)x_1 + k_2x_2 - (b_1+b_2)x_3 + b_2x_4)$$
$$\dot{x}_4 = \frac{1}{m_2}(k_2x_1 - k_2x_2 + b_2x_3 - b_2x_4 + u)$$

State $x = [x_1 \;\; x_2 \;\; x_3 \;\; x_4]^T$

Output: $y = [\dot{x}_3 \;\; \dot{x}_4]$

Expressing this is terms of the matrix form:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

We have

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -(k_1+k_2)/m_1 & k_2/m_1 & -(b_1+b_2)/m_1 & b_2/m_1 \\ k_2/m_2 & -k_2/m_2 & b_2/m_2 & -b_2/m_2 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/m_2 \end{bmatrix}$$

$$C = \begin{bmatrix} -(k_1+k_2)/m_1 & k_2/m_1 & -(b_1+b_2)/m_1 & b_2/m_1 \\ k_2/m_2 & -k_2/m_2 & b_2/m_2 & -b_2/m_2 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 \\ 1/m_2 \end{bmatrix}$$

Example 2 Model parameters

Model evaluated using the following mass, damping, and stiffness parameters.

$$m_1 = 5\ kg$$
$$m_2 = 4\ kg$$
$$b_1 = 5\ N\text{-}s/m$$
$$b_2 = 2\ N\text{-}s/m$$
$$k_1 = 150\ N/m$$
$$k_2 = 100\ N/m$$

Example 2 Is it controllable?

Given the parameters of mass, damping, and stiffness, (A, B) *are*

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -50 & 20 & -1.4 & 0.4 \\ 25 & -25 & 0.5 & -0.5 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.25 \end{bmatrix}$$

Controllability Test $\quad Q = C_{ab} = [B \quad AB \quad A^2B \quad A^3B]$

$rank(C_{ab}) = 4 = n$ Or $Det(C_{ab}) \neq 0$

Yes - system is controllable

We can design a state-feedback control assuming full state is available.

Given that the system is controllable, it means we can design and implement an Active Mass Damping control.

Example 2 Control Specifications

Control specifications: When a disturbance is applied to the structure, we want the structure to settle in 2.5 sec and have a percent overshoot less than 5% (approximately).

Overshoot =5% of 2nd order system:

$$\zeta = \frac{-\ln(OS)}{\sqrt{\pi^2 + \ln^2(OS)}}$$

$T_s = 2.5\ sec$ settling time of 2nd order system:

$$\omega_n = \frac{4}{\zeta\, T_s}$$

$$\zeta = 0.690,\ \omega_n = 2.32$$

Example 2 Desired Poles

Given (ω_n, ζ) from the second-order specifications, the two dominant poles can be located at:

$$p_1 = -\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} = -1.60 + j1.68$$

$$p_2 = -\zeta\omega_n - j\omega_n\sqrt{1-\zeta^2} = -1.60 - j1.68$$

The two other poles are then placed along the real-axis to the left of the dominant poles: (5 times)

$$p_3 = -15$$

$$p_4 = -30$$

The **desired characteristic equation** is

$$(s + 1.6 - j1.68)(s + 1.60 + j1.68)(s + 15)(s + 30)$$

$$= s^4 + 48.2s^3 + 599.4s^2 + 1681.9s + 2418.9$$

Another way:

2nd system: $s^2 + 2\zeta\omega_n s + \omega_n^2$

The desired characteristic equation is

$$(s^2 + 2\zeta\omega_n s + \omega_n^2)(s + 15)(s + 30)$$

Example 2 Pole-Placement Control

STEP 1: Find the companion matrices $(\boldsymbol{A_c}, \boldsymbol{B_c})$ of the system. Find the transformation matrix $\boldsymbol{W=Q\,Q_c^{-1}}$

STEP 2: Apply control law to the companion matrices and compute the companion control gain K_c that assigns the poles to the desired locations in the closed-loop system $(A_c - B_c\,K_c)$.

STEP 3: Find $K=K_c\,W^{-1}$ to get the feedback gain for the original system (A,B).

Example 2 Compute the control gain

Given the parameters of mass, damping, and stiffness, (A, B) *are*

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -50 & 20 & -1.4 & 0.4 \\ 25 & -25 & 0.5 & -0.5 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.25 \end{bmatrix}$$

Find the gain $K = [k_1\ k_2\ k_3\ \ k_4]$ to equate the coefficients of $det(s\boldsymbol{I} - (A - BK))$ and the Characteristic equation of system: $= s^4 + 48.2s^3 + 599.4s^2 + 1681.9s + 2418.9$

$$(A - BK) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -50 & 20 & -1.4 & 0.4 \\ 25 - 0.25k_1 & -25 - 0.25k_2 & 0.5 - 0.25k_3 & -0.5 - 0.25k_4 \end{bmatrix}$$

It is difficult to calculation. So, we do the conversion.

Example 2 Step 1) Find Controllable Companion Matrices of System

Find the companion matrices (A_c, B_c) of the system.
Find the transformation matrix $W = Q\ Q_c^{-1}$ (needed later).

Given a set of mass, damping, and stiffness parameters (A, B) are:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -50 & 20 & -1.4 & 0.4 \\ 25 & -25 & 0.5 & -0.5 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.25 \end{bmatrix}$$

Characteristic equation of system is $det(sI - A)$

$$s^4 + 1.9s^3 + 75.5s^2 + 40s + 750$$

Companion matrices:

$$A_c = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -750 & -40 & -75 & -1.9 \end{bmatrix} \quad B_c = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Controllable Matrix:

$$Q = [B \ \ AB \ \ A^2B \ \ A^3B]$$
$$Q_c = [B_c \ \ A_cB_c \ \ A_c^2B_c \ \ A_c^3B_c]$$

Transformation matrix found using controllability matrix of (A, B) and (A_c, B_c):

$$W = QQ_c^{-1} = \begin{bmatrix} 5 & 0.1 & 0 & 0 \\ 12.5 & 0.35 & 0.25 & 0 \\ 0 & 5 & 0.1 & 0 \\ 0 & 12.5 & 0.35 & 0.25 \end{bmatrix}$$

Example 2 Step 2) Compute the companion control gain K_c

Apply control law to the controllable companion system with the gain $K_c = [k_1 \;\; k_2 \;\; k_3 \;\; k_4]$ that assigns the poles to the desired locations in the closed-loop system $(A_c - B_c K_c)$.

$$A_c - B_c K_c = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -750 - k_1 & -40 - k_2 & -75 - k_3 & -1.9 - k_4 \end{bmatrix}$$

Characteristic equation of closed-loop system:

$$s^4 + (1.9 + k_4)s^3 + (75.5 + k_3)s^2 + (40 + k_2)s + 750 + k_1$$

Recall the **desired** characteristic equation:

$$s^4 + 48.2s^3 + 599.4s^2 + 1681.9s + 2418.9$$

Gains needed for system characteristic equation to match desired one:

$$k_1 + 750 = 2418.9$$

$$k_2 + 40 = 1681.9$$

$$k_3 + 75.5 = 599.4$$

$$k_4 + 1.9 = 48.2$$

$$K_c = [1668.9 \quad 1641.9 \quad 523.9 \quad 46.3]$$

Example 2 Step 3) Convert Gain for Original System

Calculate $K = K_c\, W^{-1}$ to get feedback gain for original system (A, B).

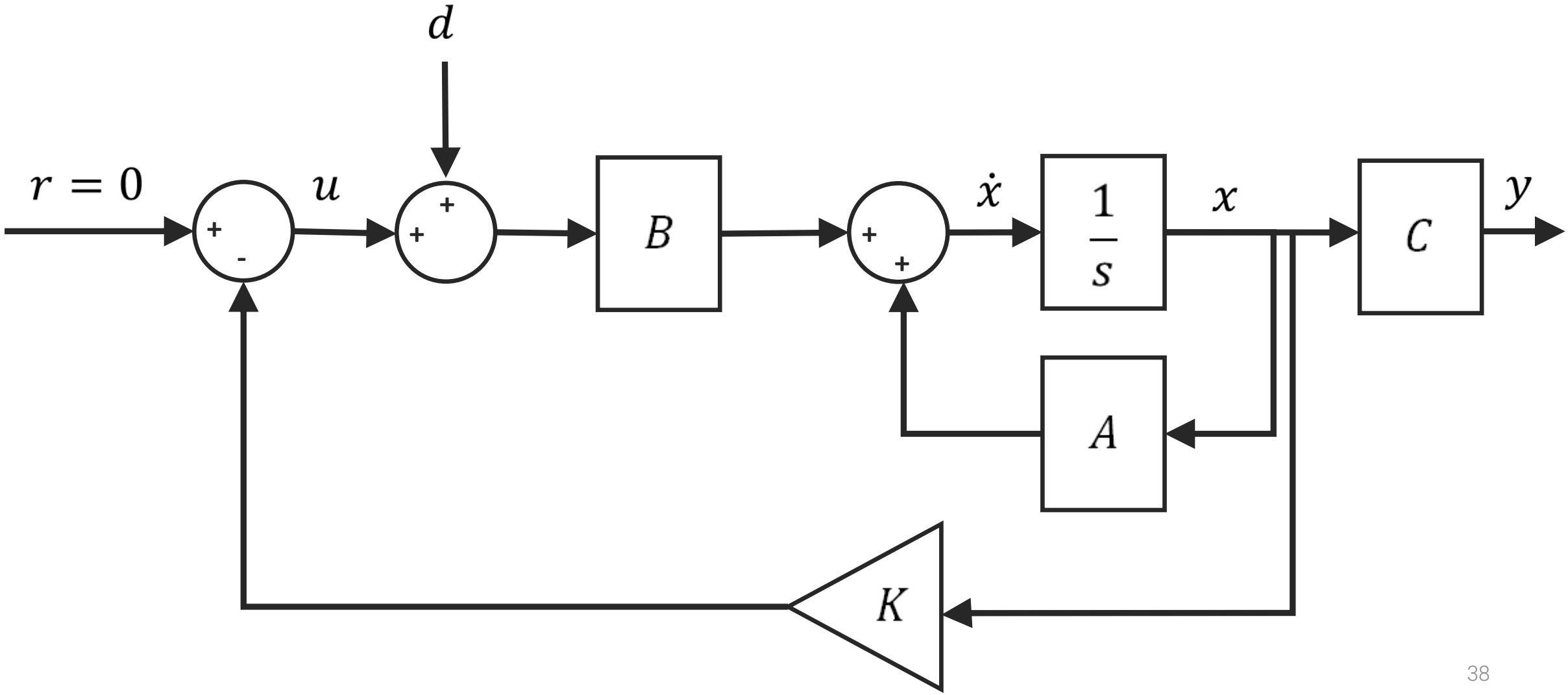
$$K_c = [1668.9 \quad 1641.9 \quad 523.9 \quad 46.3]$$

$$W = \begin{bmatrix} 5 & 0.1 & 0 & 0 \\ 12.5 & 0.35 & 0.25 & 0 \\ 0 & 5 & 0.1 & 0 \\ 0 & 12.5 & 0.35 & 0.25 \end{bmatrix}$$

$$K = K_c\, W^{-1} = [-4436.2 \quad 1908.0 \quad -179.5 \quad 185.2]$$

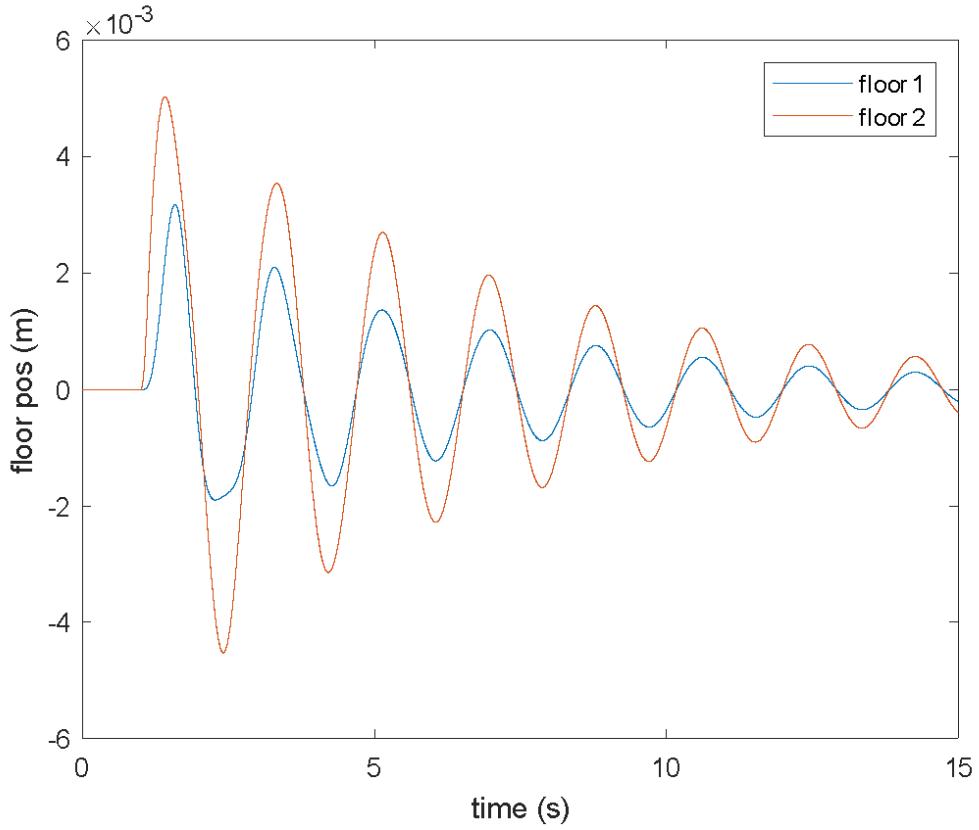
```
% Matlab
% pole placement control
% P=[p1 p2 p3 p4] are  desired poles
% A, B, C, D system state-space matrices
K=place(A,B,P)
```

State-Feedback Control with Disturbance

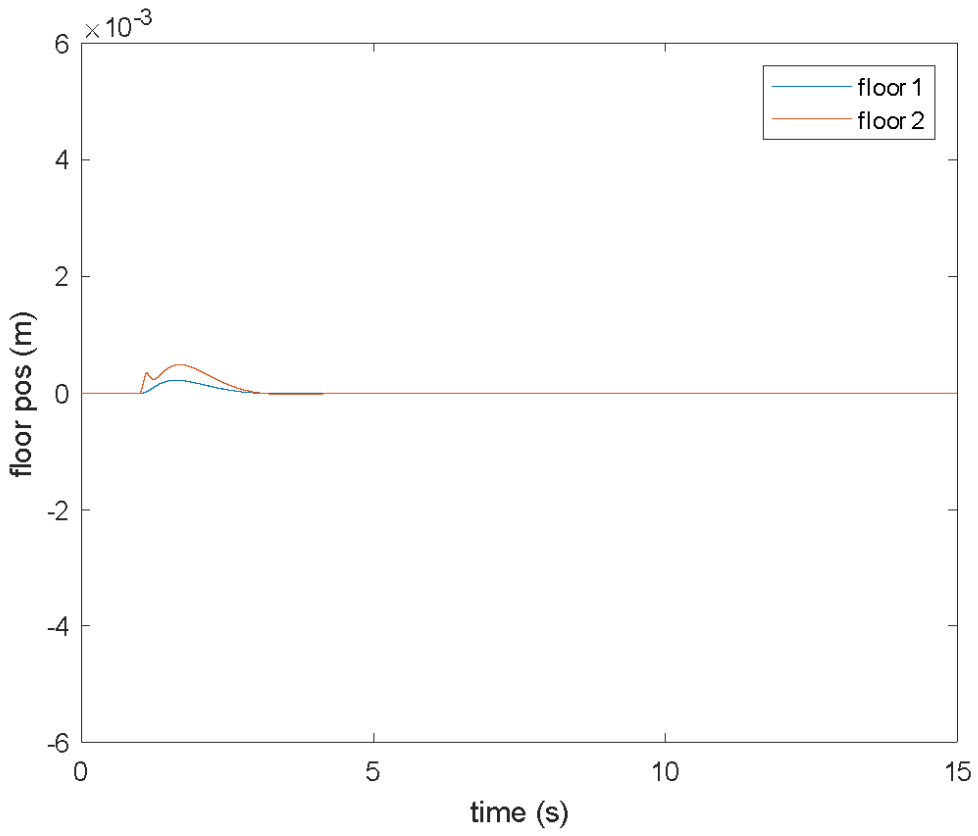


Response to 1N Disturbance on Structure

Without State-Feedback Control



With State-Feedback Control



Takeaways

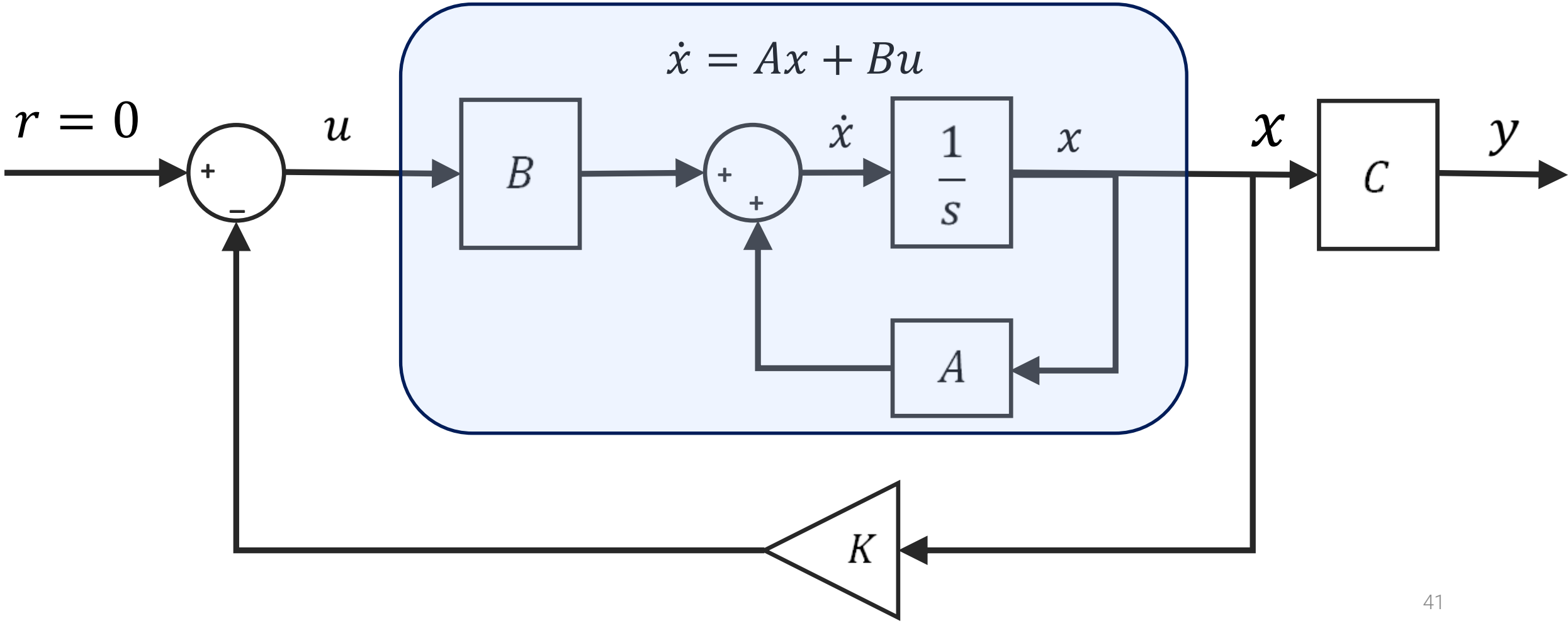
Benefits

- Design is based on time-domain/performance-based specifications
  - e.g. moving poles to a location in order to get a certain response
- Can require less control design iterations (compared to other methods)

Drawbacks

- Can have large control effort
  - Optimal control is more efficient

State Feedback $u = -Kx$



Pole Placement Method

Recall the state-space model of linear system:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

where:

A = system matrix $(n \times n)$
B = input matrix $(n \times r)$
C = output matrix $(m \times n)$
D = feedforward matrix $(m \times r)$

The system state is

$$x = [x_1 \quad x_2 \quad x_3 \ldots x_{n-1} \quad x_n]^T$$

STEP 1: Check the system is controllable. $\dot{x} = Ax + Bu$

$$C_{ab} = (B \quad AB \quad A^2B \quad \cdots \quad A^{n-1}B),\ rank(C_{ab}) = n.$$

STEP 2: Apply control law $\boldsymbol{u = -Kx}$ to the system and compute the control gain K that assigns the poles to the desired locations in the closed-loop system $(A - BK)$.

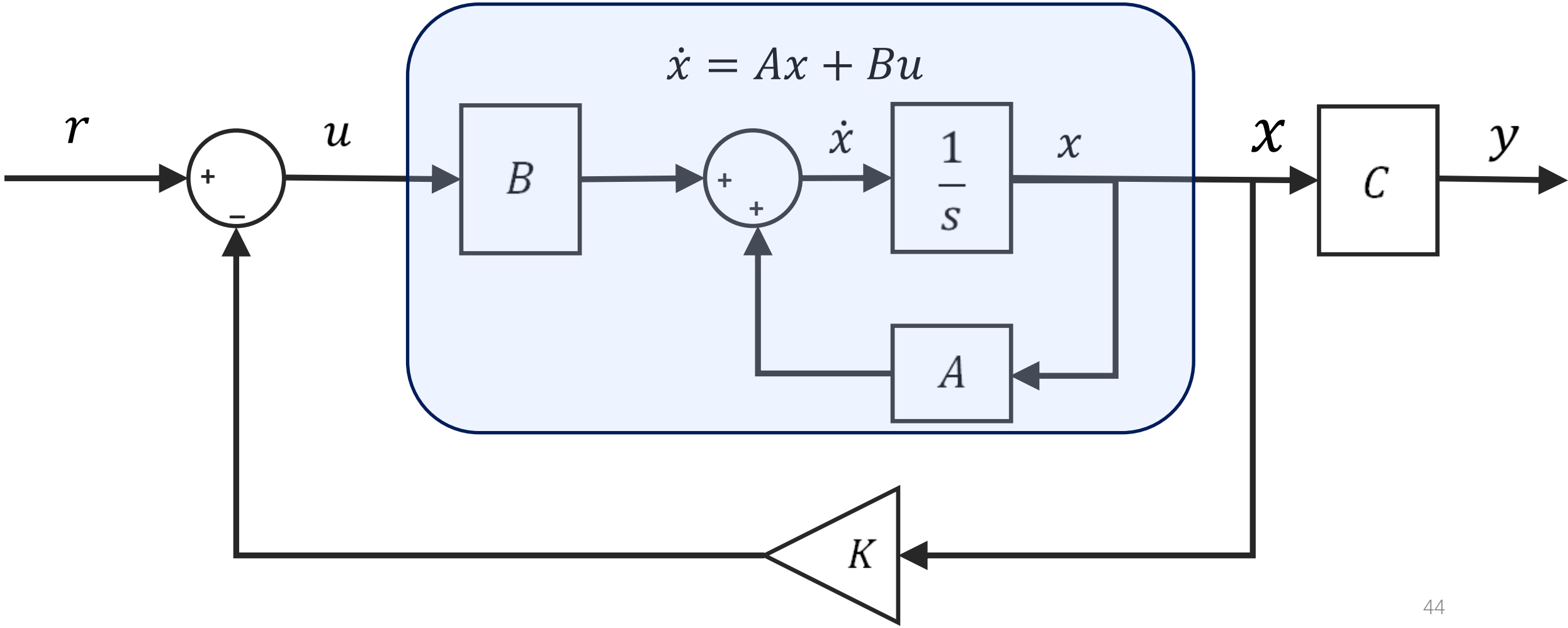
Find the gain $K = [k_1\ k_2 \cdots k_n]$ to equate the coefficients of $det(s\boldsymbol{I} - (A - BK))$ and $(s - p_1)(s - p_2) \cdots (s - p_n)$.

Pre-Filter and Pole Placement for

Reference Tracking $\lim_{\{t\to\infty\}} y(t) = Reference = Constant$

State-Feedback Control with reference ≠ 0

State-feedback control shown below can only be used if a system is controllable.



State-Feedback Control by pole placement

- *State-Space model:*

$\dot{x} = Ax + Bu$

$y = Cx$

- State-feedback control:

$u = -Kx + r$

- *Closed-loop system:*

$\dot{x} = Ax + Bu = Ax + B(-Kx + r) = (A - BK)x + Br$

$y = Cx$

- Find K using pole placement: to match the coefficients of $\mathbf{det}(\boldsymbol{sI} - (\boldsymbol{A} - \boldsymbol{BK}))$ with the coefficients of the desired characteristic polynomial (desired poles).

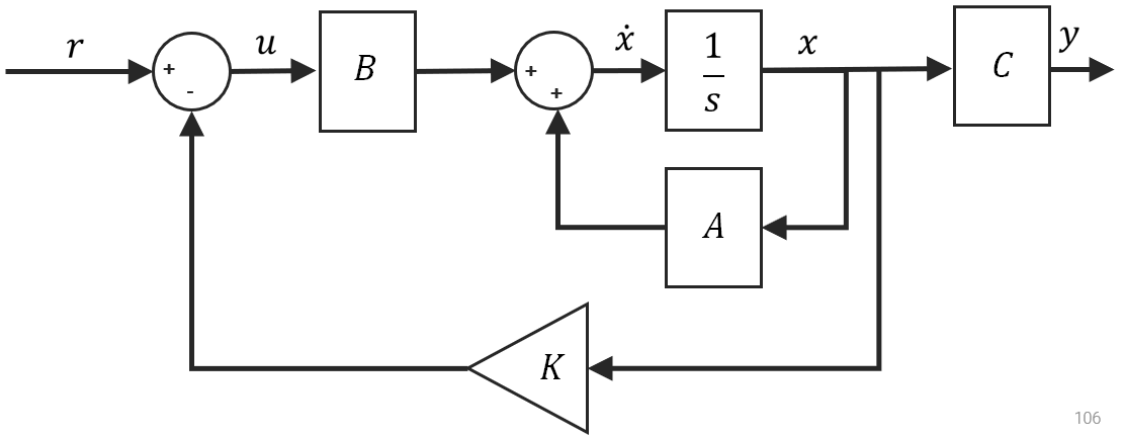
State-Feedback Control by pole placement

- *State-Space model:*

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

- State-feedback control:

$$u = -Kx + r$$



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- Find K using pole placement: to match the coefficients of $\mathbf{det}(\boldsymbol{sI} - (\boldsymbol{A} - \boldsymbol{BK}))$ with the coefficients of the desired characteristic polynomial (desired poles).

- So far, we have looked at how to pick $\boldsymbol{K}$ to get the dynamics to have some nice properties (i.e. stabilize, transient performance.)

State-Feedback Control by pole placement

- We are able to find the gain K to get the dynamics to have some nice properties (i.e. stabilization, desired poles) using pole placement.
- The question remains as to how well this controller allows us to track a reference command?
- Performance issue rather than just stability.
- For good tracking performance, we want $\boldsymbol{y(t) \rightarrow r \text{ as } t \rightarrow \infty}$.

State-Feedback Control with constant reference

State-Space model: $\dot{x} = Ax + Bu\ ,\ y = Cx$

- State-feedback control: $u = -Kx + r$

Closed-loop system:

$$\dot{x} = Ax + Bu = Ax + B(-Kx + r) = (A - BK)x + Br$$
$$y = Cx$$

- For good tracking performance we want $\lim_{\{t\to\infty\}} y(t) = r$.

Use the final value theorem: $\lim_{\{t\to\infty\}} y(t) = \lim_{\{s\to 0\}} sY(s) = \lim_{\{s\to 0\}} sR(s)$

Thus, we have $\boldsymbol{\lim_{\{s\to 0\}} \frac{Y(s)}{R(s)} = 1.}$

- So, for good performance, the transfer function from $R(s)$ to $Y\ (s)$ should be approximately 1 at DC.
- **DC gain** is the ratio of the magnitude of the steady-state step response to the magnitude of step input

Example # 3

Example # 3: Consider:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$\mathrm{y} = [1 \quad 0]\mathrm{x}$$

- Define $u = -\begin{bmatrix} k_1 & k_2 \end{bmatrix} \mathbf{x}(t) = -K\mathbf{x}(t)$, then put the poles at $s = -5, \quad -6$

Example # 3 continued

- Check the controllability of the system.

- Then $\det(sI - A) = (s-1)(s-2) - 1 = s^2 - 3s + 1 = 0$ so the system is unstable.

- Define $u = -\begin{bmatrix} k_1 & k_2 \end{bmatrix} \mathbf{x}(t) = -K\mathbf{x}(t)$, then

$$A_{cl} = A - BK = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} k_1 & k_2 \end{bmatrix} = \begin{bmatrix} 1-k_1 & 1-k_2 \\ 1 & 2 \end{bmatrix}$$

which gives

$$\det(sI - A_{cl}) = \begin{vmatrix} s-1+k_1 & k_2-1 \\ -1 & s-2 \end{vmatrix} = s^2 + (k_1 - 3)s + (1 - 2k_1 + k_2) = 0$$

Example # 3 continued

- To put the poles at $s = -5, \ -6$, compare the *desired characteristic equation*

$$(s+5)(s+6) = s^2 + 11s + 30 = 0$$

with the closed-loop one

$$s^2 + (k_1 - 3)s + (1 - 2k_1 + k_2) = 0$$

to conclude that

$$\left.\begin{array}{r} k_1 - 3 = 11 \\ 1 - 2k_1 + k_2 = 30 \end{array}\right\} \begin{array}{l} k_1 = 14 \\ k_2 = 57 \end{array}$$

so that $K = \begin{bmatrix} 14 & 57 \end{bmatrix}$, which is called **Pole Placement**.

Example # 3 continued

For the system

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}(t)$$

- Already designed $K = \begin{bmatrix} 14 & 57 \end{bmatrix}$ so the closed-loop system is

$$\dot{\mathbf{x}}(t) = (A - BK)\mathbf{x}(t) + Br$$
$$y = C\mathbf{x}(t)$$

which gives the transfer function

$$\frac{Y(s)}{R(s)} = C\left(sI - (A - BK)\right)^{-1} B$$
$$= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s+13 & 56 \\ -1 & s-2 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$= \frac{s-2}{s^2 + 11s + 30}$$

Example # 3 continued

transfer function $$\frac{Y(s)}{R(s)} = \frac{s-2}{s^2+11s+30}$$

- Assume that $r(t)$ is a step, then by the FVT

$$\left.\frac{Y(s)}{R(s)}\right|_{s=0} = -\frac{2}{30} \neq 1 \text{ !!}$$

  - So our step response is quite poor!

So, the tracking performance is not achieved, $\lim_{\{t\to\infty\}} y(t) \neq r$.

Example # 3 continued

- One solution is to scale the reference input r so that

$$u = \overline{N}r - K\mathbf{x}(t) \qquad \boldsymbol{r = Constant}$$

  - $\overline{N}$ extra gain used to scale the closed-loop transfer function

- Now we have

$$\begin{aligned} \dot{\mathbf{x}}(t) &= (A - BK)\mathbf{x}(t) + B\overline{N}r \\ y &= C\mathbf{x}(t) \end{aligned}$$

so that

$$\frac{Y(s)}{R(s)} = C\left(sI - (A - BK)\right)^{-1} B\overline{N} = G_{cl}(s)\overline{N}$$

Example # 3 continued

$$\frac{Y(s)}{R(s)} = \overline{N}\,\frac{s-2}{s^2+11s+30} = 1$$

$$\left.\frac{Y(s)}{R(s)}\right|_{s=0} = -\frac{2}{30}\,\overline{N} = 1$$

- If we had made $\overline{N} = -15$, so with a step input, $\lim_{\{t\to\infty\}}(y(t)-r) = 0$

  $u = -Kx + \bar{N}r = [14 \;\; 57]\,x - 15\,r$

- Clearly can compute

  $\overline{N} = G_{cl}(0)^{-1}$

  $G_{cl}(0) = G_{cl}(s)|_{s=0} = C\big(sI-(A-BK)\big)^{-1}B|_{s=0}$

- Note that this development assumed that r was constant, but it could also be used if r is a slowly time-varying command.

Check in Matlab.

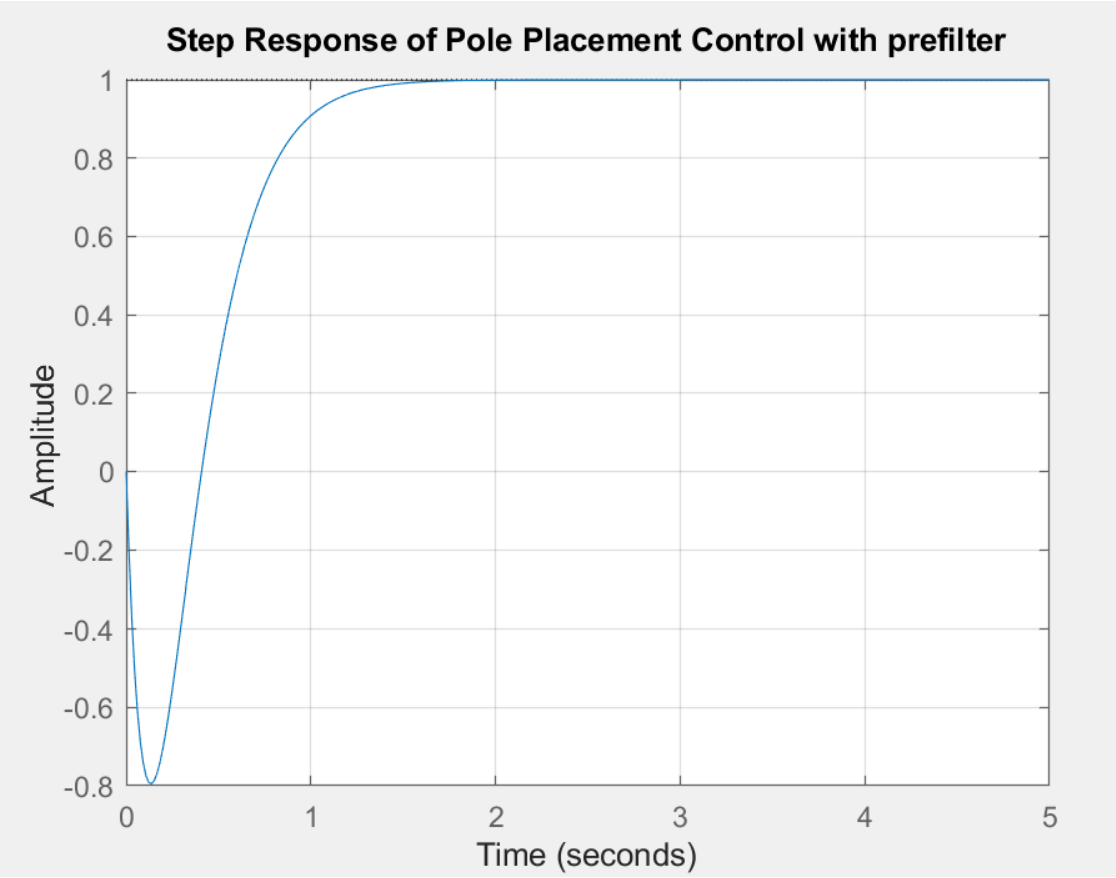
Example # 3

Consider:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

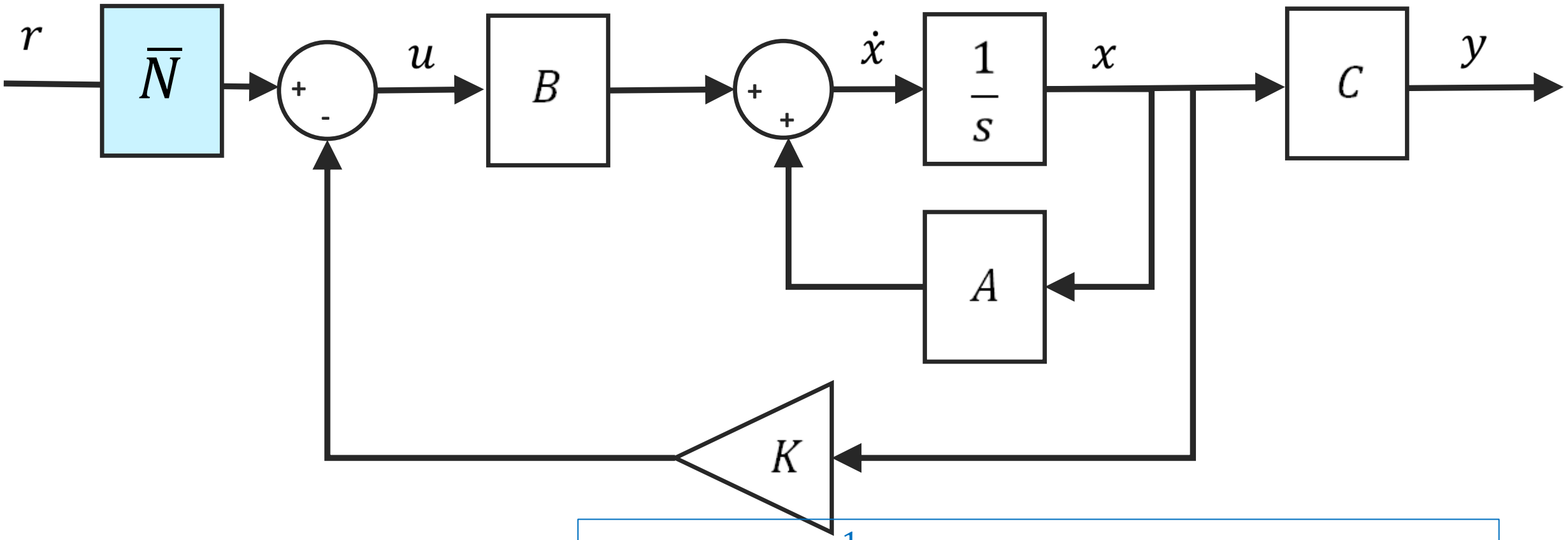
$$\mathrm{y(t)} = [1 \quad 0]\mathrm{x(t)}$$

- Design a controller u to put the poles at -5 and -6 and to track a unit step input $r(t)$.
- The control $u = -Kx + \bar{N}r$
- Using Pole placement to find the state feedback gain $K = [14 \;\; 57]$
- Using the filter to find the gain $\bar{N} = -15$
- $u = -Kx + \bar{N}r = [14 \;\; 57]\, x - 15\, r$



Matlab : K = dcgain(SYS) computes the steady-state (D.C. or low frequency) gain of the dynamic system SYS.

State-Feedback Control- Design of Pre-filter



$$u = \overline{N}r - K\mathbf{x}(t)$$

$$\overline{N} = \left(G_{cl}(0)\right)^{-1}$$

$$G_{cl}(0) = G_{cl}(s)|_{s=0} = [C\left(sI - (A - BK)\right)^{-1}B]|_{s=0}$$

$G_{cl}(0)$ is called DC gain (steady-state gain of closed-loop system)

State-Feedback Control with Pre-filter for constant reference tracking (set-point tracking)

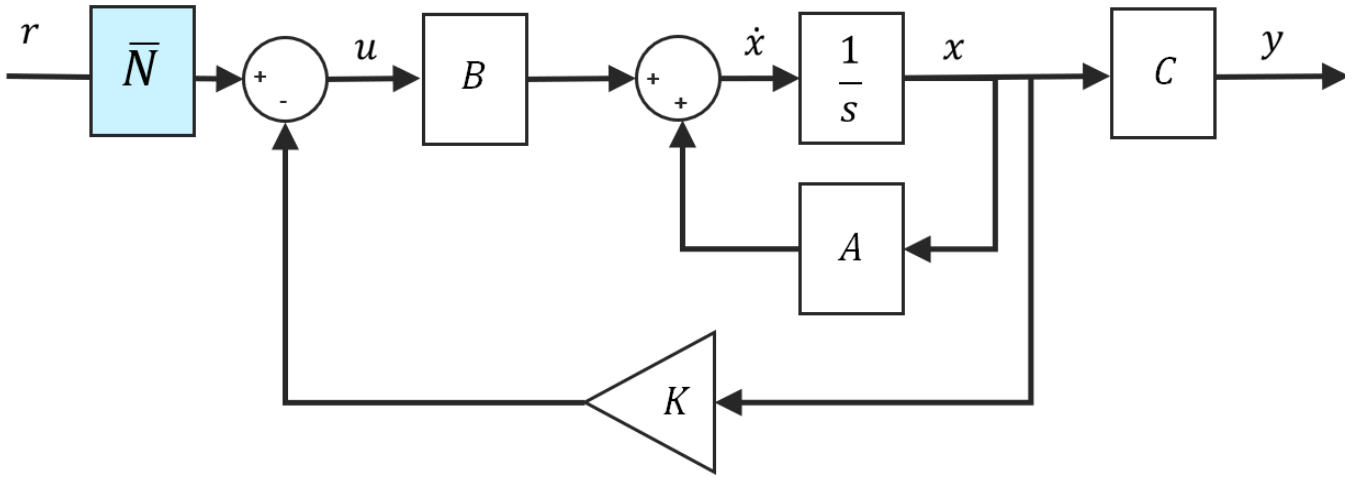
Control objective: 1) Desired poles (stability, transient performance)
2) Tracking error $= 0$, $lim_{\{t\to\infty\}}\, y(t) = r$.

- **State-feedback control**:
  $\boldsymbol{u = -Kx + \bar{N}\, r}$
- Find **K** using pole placement

- **Pre-filter**:

$\bar{N} = G_{cl}(0)^{-1}$

$G_{cl}(0) = G_{cl}(s)|_{s=0} = C\left(sI - (A - BK)\right)^{-1} B|_{s=0}$

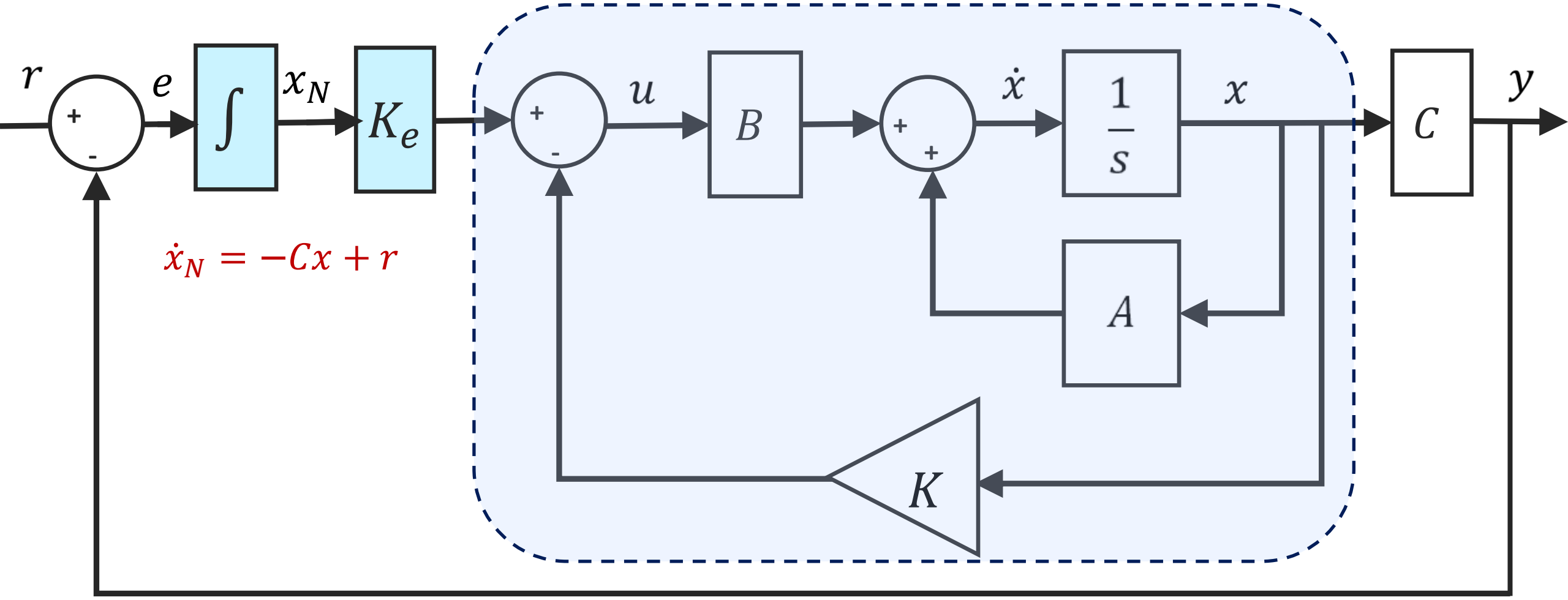


Integral Control and Pole Placement for

Constant Reference Tracking $\lim_{\{t\to\infty\}} y(t) = r$

Integral Control

$$u = -Kx + K_e x_N$$



Integral Control with reference

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

$$\dot{x}_N = -Cx + r$$

$$u = -Kx + K_e x_N$$

$$\begin{bmatrix} \dot{x} \\ \dot{x}_N \end{bmatrix} = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix} \begin{bmatrix} x \\ x_N \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u = \boldsymbol{A_I} \begin{bmatrix} x \\ x_N \end{bmatrix} + \boldsymbol{B_I} u$$

$$y = \begin{bmatrix} C & 0 \end{bmatrix} \begin{bmatrix} x \\ x_N \end{bmatrix} = \boldsymbol{C_I} \begin{bmatrix} x \\ x_N \end{bmatrix}$$

$$u = -Kx + K_e x_N = -\begin{bmatrix} K & -K_e \end{bmatrix} \begin{bmatrix} x \\ x_N \end{bmatrix}$$

The integrator increases the system type and reduces the error to zero.

We can use pole placement method to find $[K \quad -K_e]$ to the desired poles.

Note that we have an additional pole (5 times of dominant poles) to place.

Integral Control

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

$$\dot{x}_N = -Cx + r$$

$$u = -Kx + K_e x_N$$

$$\begin{bmatrix} \dot{x} \\ \dot{x}_N \end{bmatrix} = \begin{bmatrix} (A - BK) & BK_e \\ -C & 0 \end{bmatrix} \begin{bmatrix} x \\ x_N \end{bmatrix} + \begin{bmatrix} 0 \\ I \end{bmatrix} r \qquad \text{(A1)}$$

$$y = [C \quad 0] \begin{bmatrix} x \\ x_N \end{bmatrix}$$

$$u = -Kx + K_e x_N = [-K \quad K_e] \begin{bmatrix} x \\ x_N \end{bmatrix}$$

The integrator increases the system type and reduces the error to zero.

We can use the characteristic equation in (A1) to design K and K_e .

Note that we have an additional pole (5 times of dominant poles) to place.

Example 4

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -3 & -5 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 0]x$$

Design a controller using integral control

- to yield 10% overshoot and a settling time of 0.5 second and
- to track a unit step input.

Example 4 - Solution Control Specifications for 2nd order system

OS=10% overshoot of 2nd order system:

$$\zeta = \frac{-\ln(OS)}{\sqrt{\pi^2 + \ln^2(OS)}}$$

$$\zeta = 0.591,\ \omega_n = 13.533$$

$T_s = 0.5$ settling time of 2nd order system:

$$\omega_n = \frac{4}{\zeta\, T_s}$$

- The 2nd-order polynomial $s^2 + 2\zeta\omega_n\, s + \omega_n^2 = s^2 + 16s + 183.1$
- Addition pole is at -100 (5 times far from dominant poles $(-8 * 5 = -40)$
- The desired characteristic polynomial is $\boldsymbol{(s^2 + 16s + 183.1)(s + 100)}$

Example 4 - Solution

$$A=\begin{bmatrix}0 & 1\\ -3 & -5\end{bmatrix}, B=\begin{bmatrix}0\\1\end{bmatrix}, C=[1 \quad 0], K=[k_1 \quad k_2]$$

$$\begin{bmatrix}\dot{x}\\ \dot{x}_N\end{bmatrix}=\begin{bmatrix}(A-BK) & BK_e\\ -C & 0\end{bmatrix}\begin{bmatrix}x\\ x_N\end{bmatrix}+\begin{bmatrix}0_{n*1}\\ 1\end{bmatrix}r, \qquad y=[C \quad 0]\begin{bmatrix}x\\ x_N\end{bmatrix}$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_N\end{bmatrix}=\begin{bmatrix}\left(\begin{bmatrix}0 & 1\\ -3 & -5\end{bmatrix}-\begin{bmatrix}0\\1\end{bmatrix}[k_1 \quad k_2]\right) & \begin{bmatrix}0\\1\end{bmatrix}K_e\\ [-1 \quad 0] & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_N\end{bmatrix}+\begin{bmatrix}0\\0\\1\end{bmatrix}r \qquad y=[1 \quad 0 \quad 0]\begin{bmatrix}x_1\\ x_2\\ x_N\end{bmatrix}$$

$$=\begin{bmatrix}0 & 1 & 0\\ -(3+k_1) & -(5+k_2) & K_e\\ -1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_N\end{bmatrix}+\begin{bmatrix}0\\0\\1\end{bmatrix}r=A_e\begin{bmatrix}x_1\\ x_2\\ x_N\end{bmatrix}+B_e r$$

Exercises.

- Find the gain $K=[k_1 \; k_2]$ and K_e to equivalating two characteristic equations $\det(SI-A_e)$ and $(s^2+16s+183.1)(s+100)$.

Example 4 - Solution ($K = [1780,\ 111],\ K_e = 18314$)

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_N\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ -(3+k_1) & -(5+k_2) & K_e\\ -1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_N\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix} r = A_e \begin{bmatrix}x_1\\ x_2\\ x_N\end{bmatrix} + B_e r$$

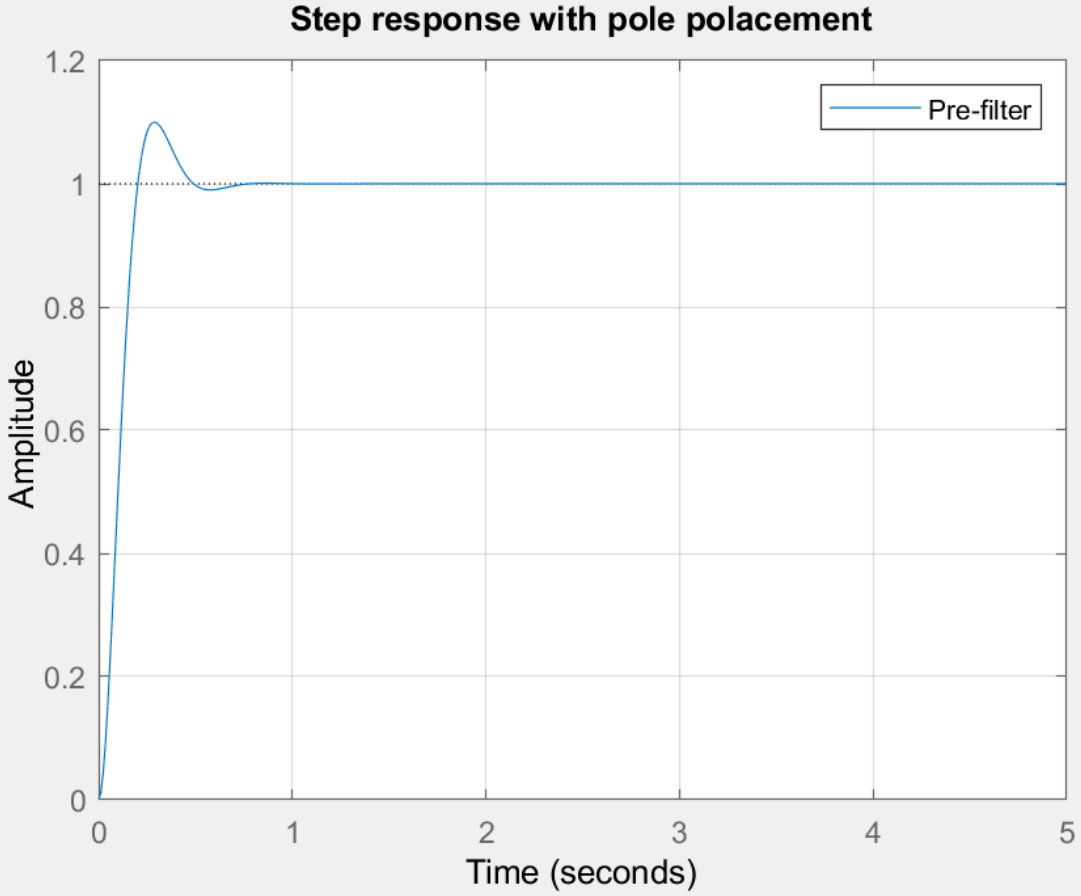
- Find the gain $K = [k_1\ k_2]$ and K_e to equivalating two characteristic equations $\det(SI - A_e)$ and $(s^2 + 16s + 183.1)(s + 100)$.

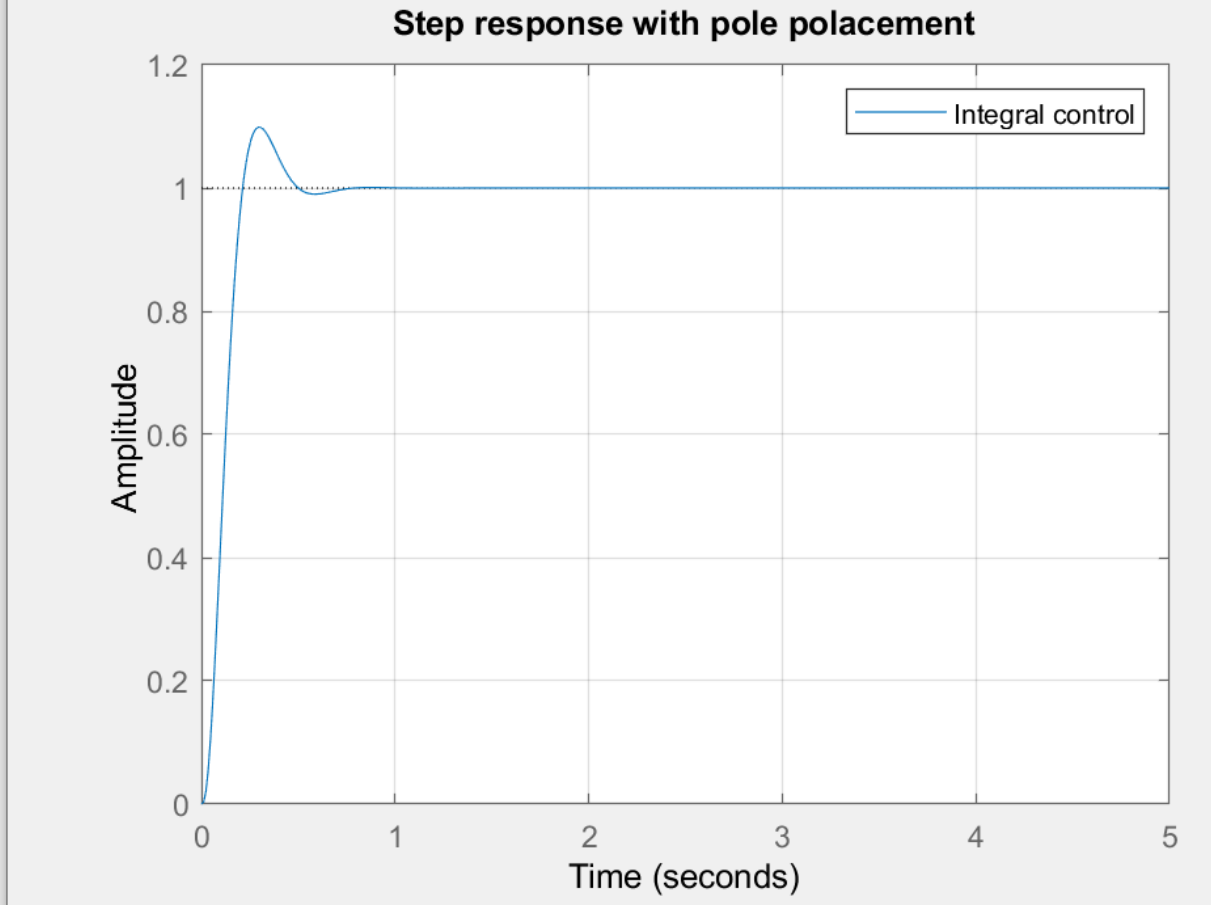
$$SI - A_e = \begin{bmatrix}s & -1 & 0\\ (3+k_1) & s+(5+k_2) & -K_e\\ 1 & 0 & s\end{bmatrix} = K_e + (3 + k_1)s + (k_2 + 5)s^2 + s^3$$

$$= (s^2 + 16s + 183.1)(s + 100) = 18314 + 1783.1s + 116s^2 + s^3$$

$$K = [k_1 \quad k_2] = [1780 \quad 111], \qquad K_e = 18314$$

Example 4 – Compare Prefilter vs. Integral Contorl





Matlab: example4_integralcontrol.m

Pole placement for partial-controllable system

Example 5. Pole Placement

Consider the following system state equation

$$\dot{x} = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} u$$

If possible, to design a feedback control $u = -kx$ to place the closed-loop poles at

- **Case I:** $-2,\ -1+j,\ -1-j,\ -3$
- **Case II:** $-2,\ -1+j,\ -1-j,\ -1$

Example 5. - continued

- **Solution.** Check the controllability matrix

$$C_{ab} = [B \quad AB \quad A^2B \quad A^3B] = \begin{bmatrix} 0 & 1 & 4 & 12 \\ 1 & 2 & 4 & 8 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

rank($\boldsymbol{C_{ab}}$)$= 3 < 4,$ the system is uncontrollable.

- Introduce the transformation matrix $\boldsymbol{P}$ for controllability decomposition.

$$P^{-1} = [B \quad AB \quad A^2B \quad q_1] = \begin{bmatrix} 0 & 1 & 4 & 0 \\ 1 & 2 & 4 & 0 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 0 \end{bmatrix}$$

where the first 3 columns are any 3 *independent* columns in $\boldsymbol{C_{ab}}$, and the remaining column $\boldsymbol{q_1}$ is selected to make $\boldsymbol{P^{-1}}$ nonsingular.

Example 5. - continued

$$P^{-1} = \begin{bmatrix} 0 & 1 & 4 & 0 \\ 1 & 2 & 4 & 0 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 0 \end{bmatrix} \qquad P = \begin{bmatrix} -2/3 & 5/9 & 0 & 4/9 \\ -1/3 & 4/9 & 0 & -4/9 \\ 1/3 & -1/9 & 0 & 1/9 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

- Then the equivalence transformation $\bar{x} = Px$ transforms

$$\dot{\bar{x}} = P\,\dot{x} = P(Ax + Bu) = PAP^{-1}\,(Px) + PB\,u = \bar{A}\,\bar{x} + \bar{B}\,u$$

- Compute $\bar{A}, \bar{B}$

$$\bar{A} = PAP^{-1} = \left[\begin{array}{ccc:c} 0 & 0 & -4 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ \hdashline 0 & 0 & 0 & -1 \end{array}\right] = \begin{bmatrix} \bar{A}_{11} & \bar{A}_{12} \\ 0 & \bar{A}_{22} \end{bmatrix} \quad \bar{B} = PB = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \bar{B}_{11} \\ 0 \end{bmatrix}$$

Example 5. - continued

- Then the equivalence transformation $\bar{x} = Px$ transforms

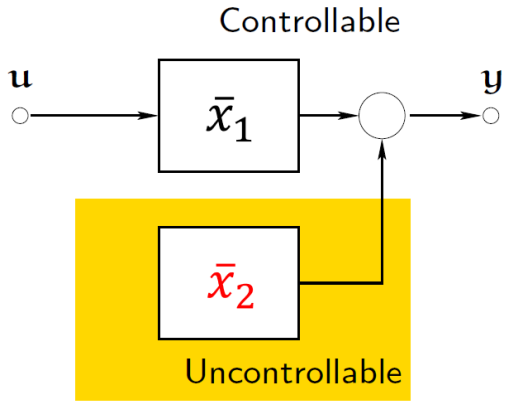
$$\dot{\bar{x}} = P\,\dot{x} = P(Ax + Bu) = PAP^{-1}\,(Px) + PB\,u = \bar{A}\,\bar{x} + \bar{B}\,u$$

$$\dot{\bar{x}} = \begin{bmatrix} \dot{\bar{x}}_1 \\ \dot{\bar{x}}_2 \end{bmatrix} = \bar{A}\,\bar{x} + \bar{B}\,u = \begin{bmatrix} \bar{A}_{11} & \bar{A}_{12} \\ 0 & \bar{A}_{22} \end{bmatrix} \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} + \begin{bmatrix} \bar{B}_1 \\ 0 \end{bmatrix} u$$

- The states in the new coordinates are decomposed into

$\bar{x}_1$: 3 *controllable states*

$\bar{x}_2$: 1 *uncontrollable state*



Example 5. - continued

$$\bar{\mathrm{A}} = \mathrm{PAP}^{-1} = \begin{bmatrix} 0 & 0 & -4 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & \mathbf{-1} \end{bmatrix} = \begin{bmatrix} \bar{A}_{11} & \bar{A}_{12} \\ 0 & \bar{A}_{22} \end{bmatrix} \quad \bar{B} = \mathrm{P}B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \bar{B}_{11} \\ 0 \end{bmatrix}$$

- This indicates there is one eigenvalue (-1) of $\boldsymbol{A}$ cannot be altered no matter how the control gain $\mathbf{k}$ is chosen.

- In other words, there is always a closed-loop located at -1 no matter how $\boldsymbol{k}$ in feedback control $\mathrm{u} = -\mathbf{kx}$ is chosen.



- **Case I:** It is ***impossible*** to find the gain k in $\mathrm{u} = -\mathbf{kx}$ *to place the closed-loop poles at* $-2, -1+j, -1-j, -3$

- **Case II:** $-2, -1+j, -1-j, -1$. It is ok.

Example 5. - continued

- **Case II:** To find a $\mathbf{k}$ to place the closed-loop poles at $-2, -1-j, -1+j, -1$. Firstly, let $\bar{\mathbf{k}} = [\bar{k}_0, \bar{k}_1, \bar{k}_2, \bar{k}_3] = [\bar{\mathbf{k}}' \ \bar{k}_3]$

- The characteristic polynomial of matrix $(\bar{\mathbf{A}} - \bar{\mathrm{B}}\bar{\mathbf{k}})$ is

$$\det[sI - (\bar{A} - \bar{B}\bar{\mathbf{k}})]$$
$$= \det[sI - (\bar{A}_{11} - \bar{B}_{11}\bar{\mathbf{k}}')](s+1)$$
$$= [s^3 + (\bar{k}_0 - 3)s^2 + (\bar{k}_1 - 3\bar{k}_0)s + \bar{k}_2 - 3\bar{k}_1 + 4](s+1)$$

- It is same to find a $\bar{\mathbf{k}}' = [\bar{k}_0, \bar{k}_1, \bar{k}_2]$ for controllable subsystem $(\bar{A}_{11}, \bar{B}_{11})$ to place the closed-loop poles at $-2, -1-j, -1+j$.

Example 5. - continued

- $\bar{\mathbf{k}}' = [\bar{k}_0, \bar{k}_1, \bar{k}_2]$ for controllable subsystem $(\bar{A}_{11}, \bar{B}_{11})$ to place the closed-loop poles at $-2, -1-j, -1+j$.
- Match the closed-loop system $(\bar{A}_{11} - \bar{B}_{11}\bar{\mathbf{k}}')$ to the desired poles polynomial.

$$\bar{A}_{11} = \begin{bmatrix} 0 & 0 & -4 \\ 1 & 0 & 0 \\ 0 & 1 & 3 \end{bmatrix}, \qquad \bar{B}_{11} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Example 5. - continued

- The desired characteristic polynomial of the closed-loop system is

$$(s+2)(s+1-j)(s+1+j) = (s^3+4s^2+6s+4)$$

$$= \left[s^3 + (\bar{k}_0 - 3)s^2 + (\bar{k}_1 - 3\bar{k}_0)s + \bar{k}_2 - 3\bar{k}_1 + 4\right]$$

- Compute $\bar{k}_0, \bar{k}_1, \bar{k}_2$ by comparing the coefficients of the two characteristic polynomials

$$\begin{cases} \bar{k}_0 - 3 = 4 \\ \bar{k}_1 - 3\bar{k}_0 = 6 \\ \bar{k}_2 - 3\bar{k}_1 + 4 = 4 \end{cases} \Rightarrow \begin{cases} \bar{k}_0 = 7 \\ \bar{k}_1 = 27 \\ \bar{k}_2 = 81 \end{cases}$$

$\bar{k}_3$ can be **arbitrarily** chosen. For simplicity, choose $\bar{k}_3 = 0$.

$$\bar{k} = [7 \;\; 27 \;\; 81 \;\; 0]$$

- Finally, compute $\mathbf{k}$ $\quad k = \bar{k}P = [13.3 \quad 6.89 \quad 0 \quad 0.111]$